something that everyone would agree upon, was not fully debunked by special relativity. Absolute space does not exist. Absolute time does not exist. But according to special relativity, absolute spacetime does exist. With this observation, let's visit the bucket once again.

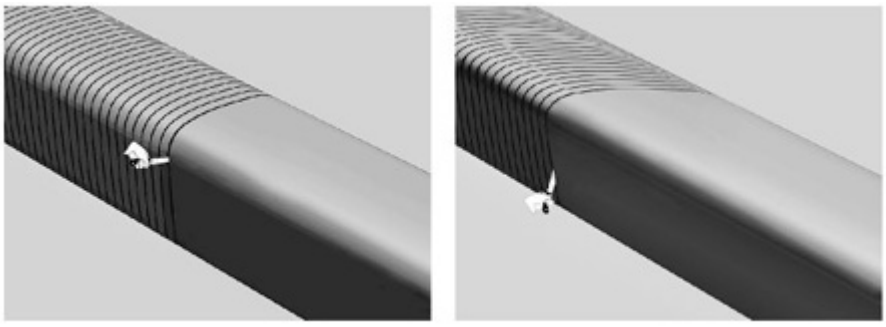


Figure 3.5 Just as one loaf of bread can be sliced at different angles, a block of spacetime is "time sliced" at different angles by observers in relative motion. The greater the relative speed, the greater the angle (with a maximum angle of 45˚ corresponding to the maximum speed set by light).

In an otherwise empty universe, with respect to *what* is the bucket spinning? According to Newton, the answer is absolute space. According to Mach, there is no sense in which the bucket can even be said to spin. According to Einstein's special relativity, the answer is absolute spacetime.

To understand this, let's look again at the proposed street and avenue layouts for Springfield. Remember that Marge and Lisa disagreed on the street and avenue address of the Kwik-E-Mart and the nuclear plant because their grids were rotated relative to each other. Even so, regardless of how each chose to lay out the grid, there are some things they definitely still agree on. For example, if in the interest of increasing worker efficiency during lunchtime, a trail is painted on the ground from the nuclear plant straight to the Kwik-E-Mart, Marge and Lisa will not agree on the streets and avenues through which the trail passes, as you can see in Figure 3.6. But they will certainly agree on the *shape* of the trail: they will agree that it is a straight line. The geometrical shape of the painted trail is independent of the particular street/avenue grid one happens to use.